

Task (1)

RNN & LSTM

Cellula Technologies

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RNN-Model

Main Dataset

Model Architecture

- Embedding Layer
- Simple RNN
- Dropout
- Dense Layer
- Softmax Output

Hyperparameters:

- Embedding Dimension: 128
- RNN Units: 128
- Dropout: 0.3–0.4
- Activation Function: tanh
- Optimizer: Adam

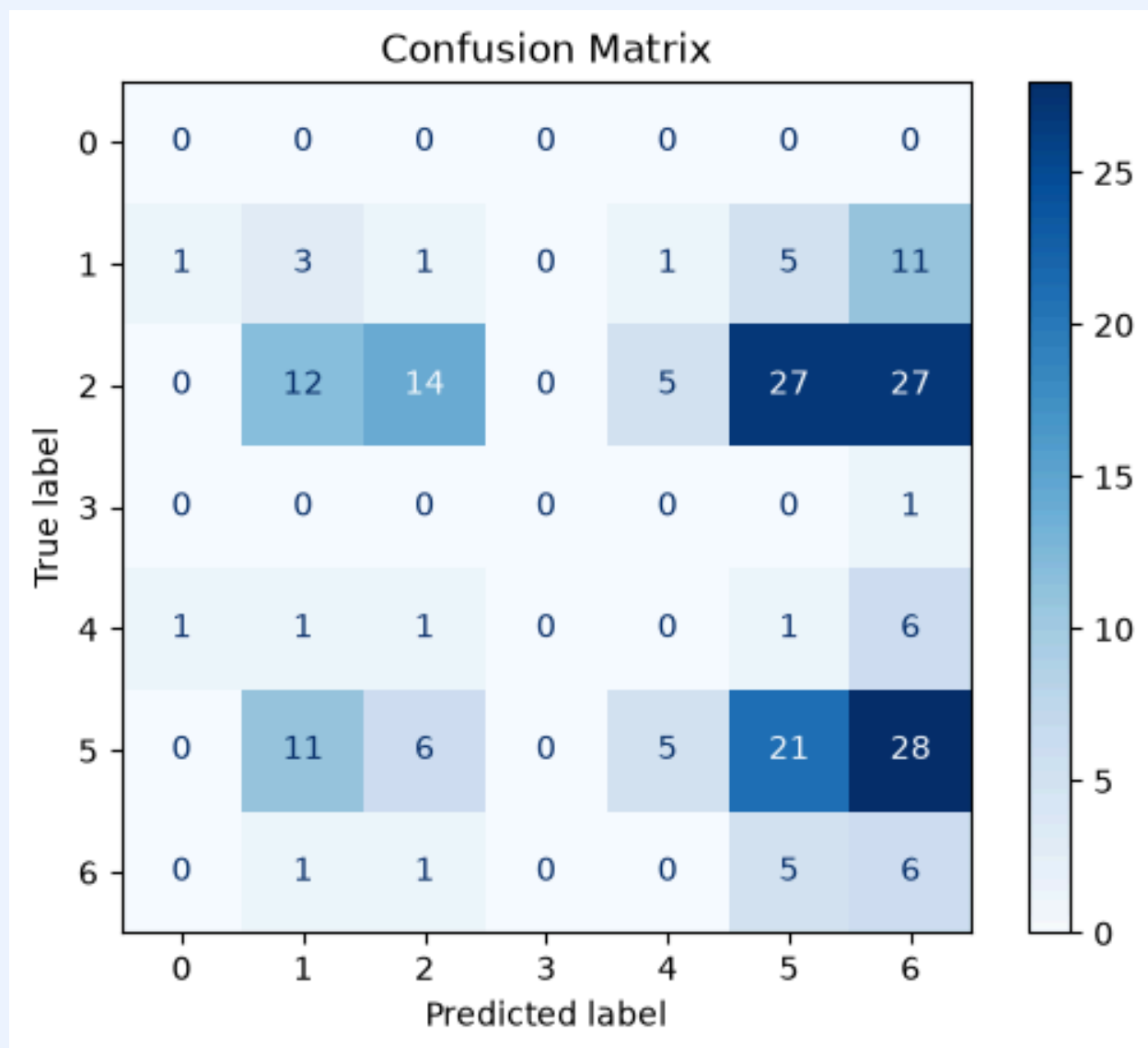
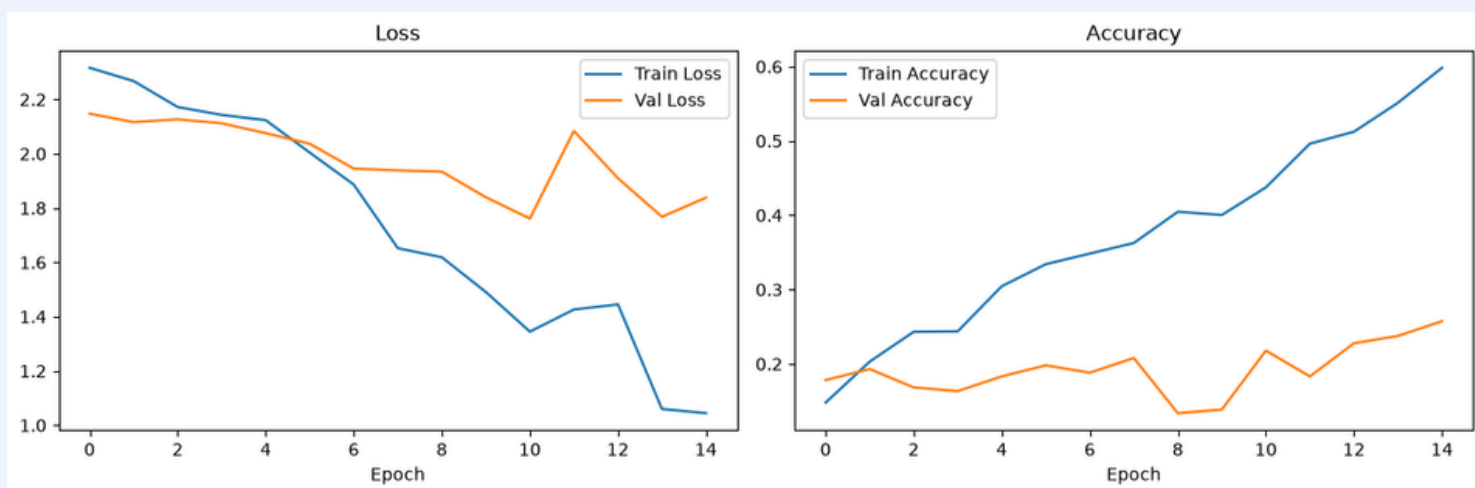
Performance:

Weighted F1-score = 0.30

The confusion matrix and classification report indicate that the model struggled to distinguish several classes, resulting in relatively low performance.

Possible reasons include:

- Small dataset.
- Large number of classes.
- Class imbalance.
- Difficulty of Vanilla RNN in learning long-term dependencies.



2nd Dataset

Model Architecture

- Embedding Layer
- Simple RNN
- Dense Layer
- Sigmoid Output

Performance

Accuracy = 90%

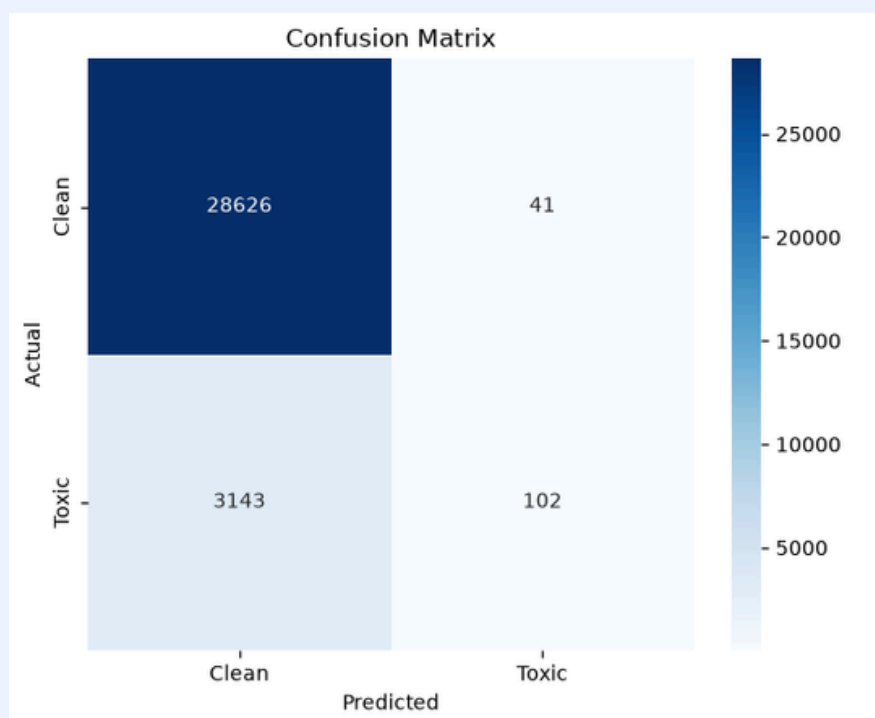
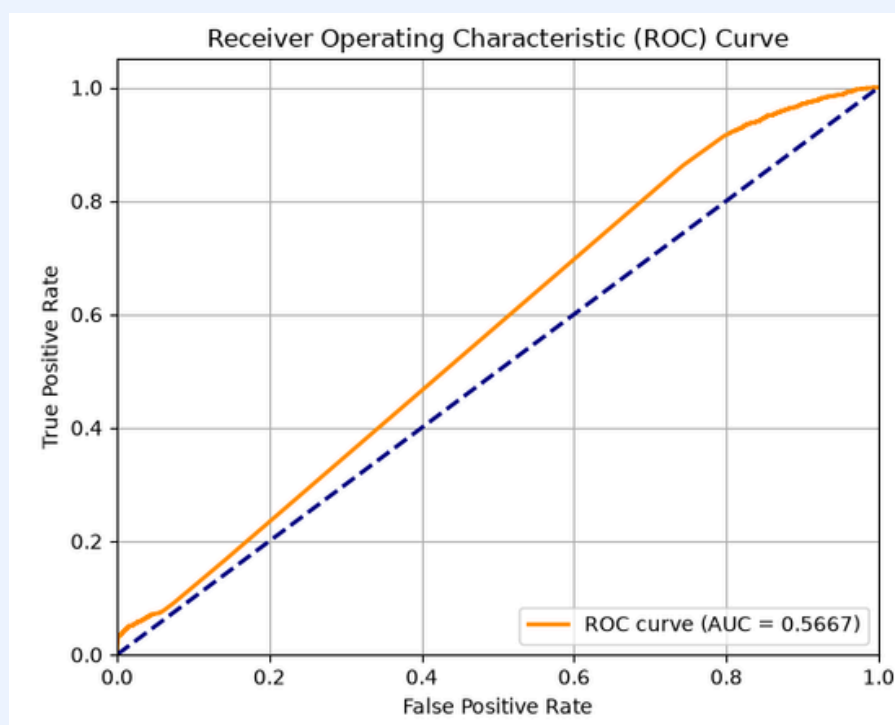
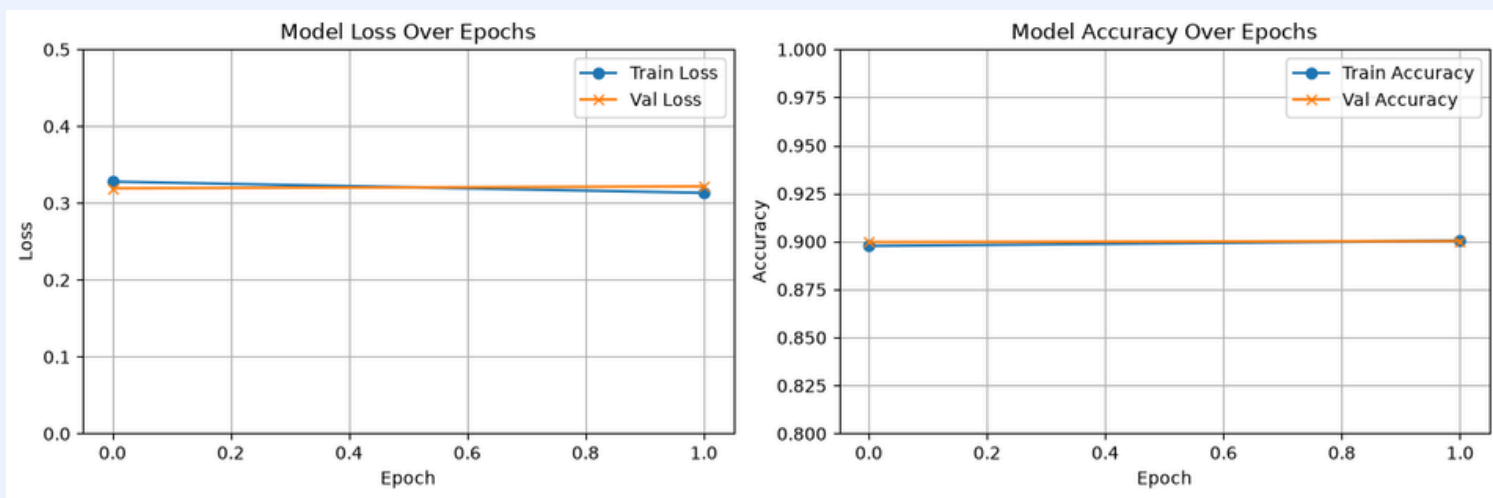
Weighted F1-score = 0.86

Although the overall accuracy is very high, the classification report reveals that the model performs much better on the Clean class than on the Toxic class.

For example:

- Clean comments:
 - Precision: 0.90
 - Recall: 1.00
- Toxic comments:
 - Precision: 0.71
 - Recall: 0.03

This indicates that the dataset suffers from class imbalance, causing the model to predict the majority class much more frequently.



LSTM Model

Model Architecture

The LSTM model consisted of:

- Embedding Layer
- Spatial Dropout
- Bidirectional LSTM
- Dropout
- Dense Layer
- Softmax Output

Additional techniques:

- L2 Regularization
- Early Stopping
- Reduce Learning Rate on Plateau

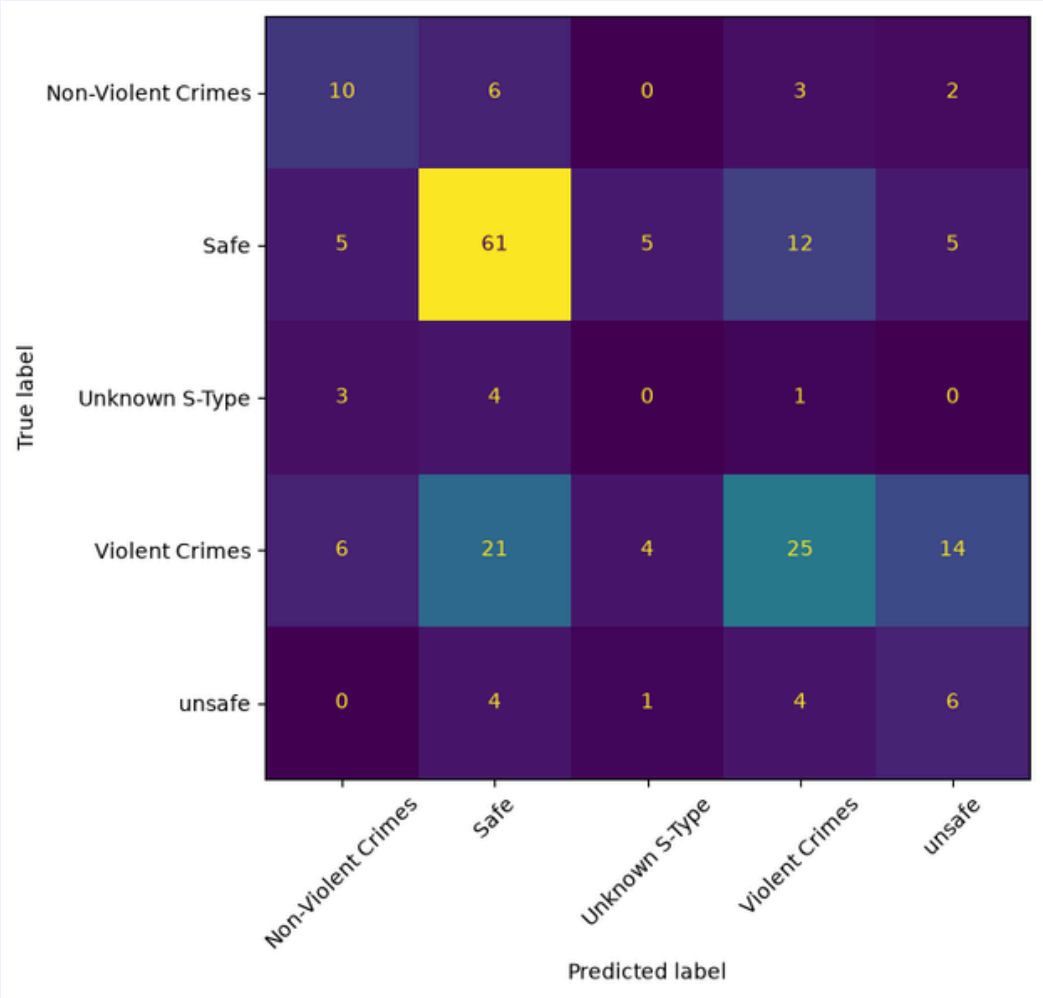
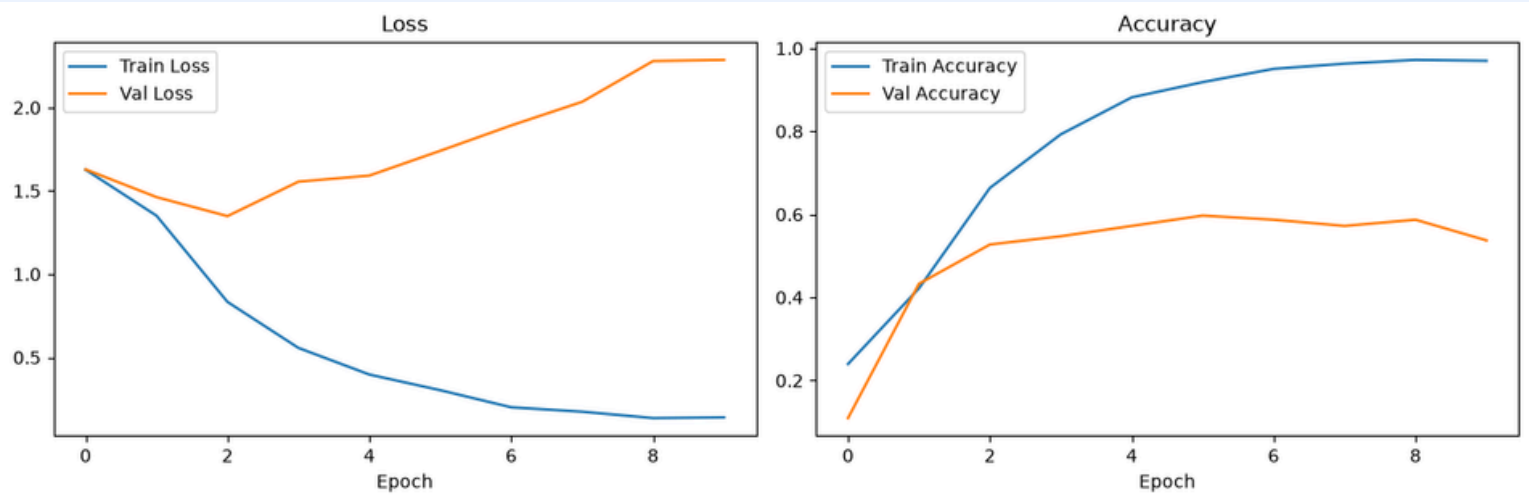
Hyperparameters

- Embedding Dimension: 64
- LSTM Units: 64
- Bidirectional LSTM
- Dropout: 0.3

Performance

Accuracy = 50%

Weighted F1-score = 0.507



Conclusion

- The comparison highlights that model performance depends not only on the neural network architecture but also on the characteristics of the dataset.
- The first RNN experiment achieved the lowest performance due to the complexity of the multi-class dataset and the limited capability of a vanilla RNN to capture long-term contextual information.
- The second RNN experiment obtained the highest overall accuracy because it was trained on a binary classification dataset. However, the evaluation metrics reveal that the model was heavily biased toward the majority class, resulting in poor recall for toxic comments. Therefore, the high accuracy alone does not fully reflect the model's effectiveness.
- The LSTM model outperformed the first RNN experiment despite solving a more challenging problem. Its ability to preserve long-term dependencies through memory cells allowed it to achieve a higher weighted F1-score and better overall class discrimination.
- Overall, the models demonstrate that LSTM networks are generally more suitable than vanilla RNNs for complex text classification tasks, especially when contextual information plays an important role.